Systematic Investment Architecture

From Signals to Strategies to Portfolios

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Outline

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Motivation

- These notes accompany the Duke Economics course *Machine Learning for Finance*.
- The course focuses on how machine learning methods can be used to design, evaluate, and implement systematic investment strategies.
- The goal of this document is to lay out a clear, modular architecture for a systematic investment process.
- Many practical considerations (e.g., transaction costs, slippage, execution details) are intentionally set aside so we can focus on the components most relevant for illustrating machine learning ideas.
- The framework is designed to be general enough to accommodate a wide range of signals, strategies, and portfolios. Throughout, a simple momentum example is used to make concepts concrete.

Q: Signal vs Strategy vs Portfolio?

A:

- Signal: tickers, direction of trade
- Strategy: weights, entry and exit timing
- Portfolio: combination of strategies that optimizes weights subject to global constraints

Portfolio Creation Process

$$X_{i,t}, Z_t \longrightarrow s_{i,t} \longrightarrow s_{i,t}^{pos} \longrightarrow w_{i,t}^{\text{raw}} \longrightarrow w_{i,t}^{\text{scaled}} \longrightarrow w_{i,t}^{p}$$

$X_{i,t}, Z_t$	$s_{i,t}$	$s_{i,t}^{pos}$	$w_{i,t}^{\text{raw}}$	$w_{i,t}^{\text{scaled}}$	$w_{i,t}^{p}$
Features	Signal	Position Timing	Position Sizing	Scaled / Constrained	Portfolio Weights

- t: highest frequency at which portfolio can be traded (e.g. end of week, end of day, intraday);
- $X_{i,t}$: characteristics of asset i at time t, e.g. returns, volume, sector
- Z_t: general features, e.g. characteristics of assets other than asset i, macro state variables
- $s_{i,t}$: signal $\in \{-1, 0, 1\}$ defining short, neutral, or long position
- $s_{i,t}^{pos}$: position $\in \{-1, 0, 1\}$ implied by signal after timing and exit rules
- $w_{i,t}^{raw}$: signed pre-normalized position sizes; e.g. proportional to signal strength
- $w_{i,t}^{scaled}$: position weights scaled to enforce strategy constraints such as fully invested
- $w_{i,t}^{p}$: optimal weighting across potentially multiple strategies and global constraints

A Toy Momentum Signal

Position sizes scaled by momentum magnitude

Stage	Definition / Rule	Comment
t	Daily (end of day)	Trading frequency
$X_{i,t}$	$mom_{i,t}$	Past-day return (momentum)
Z_t	—	No contextual variables
$s_{i,t}$	$\begin{cases} +1 & \text{if } mom_{i,t} > 0 \\ 0 & \text{if } mom_{i,t} = 0 \\ -1 & \text{if } mom_{i,t} < 0 \end{cases}$	Raw directional signal
$s_{i,t}^{\text{pos}}$	$= s_{i,t}$	Position (no timing adjustment)
$w_{i,t}^{\text{raw}}$	$= s_{i,t}^{\text{pos}} \lvert mom_{i,t} \rvert$	Size by momentum magnitude
$w_{i,t}^{\text{scaled}}$	$\frac{w_{i,t}^{\text{raw}}}{\sum_{j=1}^{N} \lvert w_{j,t}^{\text{raw}} \rvert}$	Normalize total abs. exposure
$w_{i,t}^{p}$	$= w_{i,t}^{\text{scaled}}$	Final portfolio weights

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Defining Signals

Using logical conditions for long, short, neutral positions

Define a directional signal using opposing predicates:

$$s_{i,t} = \mathbf{1}\big[p^{(+)}(X_{i,t}, Z_t)\big] - \mathbf{1}\big[p^{(-)}(X_{i,t}, Z_t)\big], \qquad s_{i,t} \in \{-1, 0, +1\}$$

- $p^{(+)}$: condition for a long signal
- $p^{(-)}$: condition for a short signal
- $s_{i,t} = +1$: long condition true, short false
- $s_{i,t} = -1$: short condition true, long false
- $s_{i,t} = 0$: neither (no signal) or both (conflict) true $\Rightarrow$ neutral

Signal A

Define mom_i as momentum on asset i [ignore the precision definition of momentum for now]

$$
\begin{aligned}
s_{i,t} =& \mathbf{1}\Big[p^{(+)}(X_{i,t}, Z_t)\Big] - \mathbf{1}\Big[p^{(-)}(X_{i,t}, Z_t)\Big] \\
p^{(+)} =& mom_{i,t-1} > 0 \\
p^{(-)} =& \; mom_{i,t-1} < 0
\end{aligned}
$$

Q: Can you explain this signal?
A:

- Long Positive Momentum, Short Negative Momentum
- $X_{i,t}$ contains $mom_{i,t-1}$
- Z_t not used
- Note: could set a small epsilon around 0 ala $mom_{i,t-1} < -\epsilon$ and $mom_{i,t-1} > \epsilon$

Signal B

Q: Can you describe this signal?

$$
\begin{aligned}
s_{i,t} =& \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right] \\
p^{(+)} =& (\mathrm{industry}_i = \text{Technology}) \\
p^{(-)} =& \text{False}
\end{aligned}
$$

A:

- Go long all stocks classified as Technology.
- Single positive condition; no explicit short rule.
- $s_{i,t} = +1$ for Technology stocks; $s_{i,t} = 0$ otherwise.
- $X_{i,t}$ includes sector/industry classification.
- Z_t not used in this example.

Signal C

Q: Can you describe this signal?

$$
\begin{aligned}
s_{i,t} =& \mathbf{1}\Big[p^{(+)}(X_{i,t}, Z_t)\Big] - \mathbf{1}\Big[p^{(-)}(X_{i,t}, Z_t)\Big] \\
p^{(+)} =& (\mathrm{industry}_i = \text{Technology}) \wedge (\mathrm{PE}_{i,t} < \overline{\mathrm{PE}}_{i,\text{hist}}) \\
p^{(-)} =& \text{False}
\end{aligned}
$$

A:

- Go long Technology stocks whose P/E is below their own historic average
- Two conditions define the long side; no short predicate.
- $X_{i,t}$: firm fundamentals and P/E history.
- Z_t not used in this example.

Signal D

Q: Can you describe this signal?

$$
\begin{aligned}
s_{i,t} =& \mathbf{1}\Big[p^{(+)}(X_{i,t}, Z_t)\Big] - \mathbf{1}\Big[p^{(-)}(X_{i,t}, Z_t)\Big] \\
p^{(+)} =& \big[\mathrm{PE}_{i,t} < \overline{\mathrm{PE}}_t\big] \\
p^{(-)} =& \big[\mathrm{PE}_{i,t} > \overline{\mathrm{PE}}_t\big]
\end{aligned}
$$

A:

- Go long stocks whose P/E is below the cross-sectional average P/E and short those above it.
- Two conjunctive conditions, one long and one short predicate
- $X_{i,t}$: asset P/E
- Z_t: cross-sectional average of P/E

Signal E

Q: Can you describe this signal?

Forecast returns via $\hat{r}_{i,t+1} = AR(2)(X_{i,t})$

$$
\begin{aligned}
s_{i,t} &= \mathbf{1}\Big[p^{(+)}(X_{i,t}, Z_t)\Big] - \mathbf{1}\Big[p^{(-)}(X_{i,t}, Z_t)\Big], \\
p^{(+)}(X_{i,t}, Z_t) &\equiv \left(\mathrm{rank}_t(\hat{r}_{i,t+1}) \leq 10\right), \\
p^{(-)}(X_{i,t}, Z_t) &\equiv \left(\mathrm{rank}_t(\hat{r}_{i,t+1}) > N_t - 10\right).
\end{aligned}
$$

Forecasts are ranked from highest to lowest, so rank 1 denotes the highest forecasted return.

A:

- Go long the 10 highest forecasts; short the 10 lowest
- Two conditions, one long and one short predicate
- $X_{i,t}$: lagged returns
- Z_t: universe size / ranking rule

Signal F

Q: Can you describe this signal?

$$s_{i,t} = \mathbf{1}\Big[p^{(+)}(X_{i,t}, Z_t)\Big] - \mathbf{1}\Big[p^{(-)}(X_{i,t}, Z_t)\Big],$$

$$p^{(+)}(X_{i,t}, Z_t) \equiv (\mathrm{industry}_i \in \{\mathsf{Tech}, \mathsf{Comm}\}) \wedge (\mathrm{mom}_{i,t-1} > \overline{\mathrm{mom}}_t) \wedge (\mathrm{VIX}_t \geq 25)$$

$$p^{(-)}(X_{i,t}, Z_t) \equiv (\mathrm{industry}_i \in \{\mathsf{Tech}, \mathsf{Comm}\}) \wedge (\mathrm{mom}_{i,t-1} > \overline{\mathrm{mom}}_t) \wedge (\mathrm{VIX}_t \leq 20)$$

A:

- Among Tech and Comm stocks with above-average momentum, go long when $\mathrm{VIX}_t \geq 25$ and short when $\mathrm{VIX}_t \leq 20$.
- Otherwise, issue no signal
- $X_{i,t}$: asset-specific information, including lagged returns and industry classification.
- Z_t: cross-sectional momentum information and the VIX.

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Sampling Danger

Q: What's the danger in evaluating this toy momentum signal?

- Construct Signal: Compute momentum on stock A for Dec2025 as cumulative return from Jan2025 through Dec2025. Repeat for stocks B,C,D,E.
- Evaluate Signal: Compare Dec2025 returns for those with positive momentum (A,B) to those with negative momentum (C,D,E).

A:

- Time series: Using Jan through Dec to evaluate a Dec trade exposes you to "look-ahead" bias; you would not have been able to place a trade at the beginning of Dec because the signal wouldn't have been known until the end of Dec [MUST AVOID]
- Cross Section: Constructing your signal on the whole universe of stocks exposes you to "in sample overfitting"; you don't know if this signal is useful for all assets or just the few you've studied [USE WITH CAUTION]

Test (aka Hold-Out) vs Train

Using Test/Train Samples to reduce the dangers in the toy momentum signal

- Time Series:
  - Train: Construct momentum on Jan through Nov
  - Test: Evaluate signal on Dec
- Cross Section:
  - Train: Construct momentum on stocks A,B,C,D,E
  - Test: Evaluate signal on stocks F,G,H
- Panel:
  - Train: Construct momentum on stocks A,B,C,D,E using Jan-Nov
  - Test: Evaluate signal on stocks F,G,H during Dec

Note: We can refer to these three as Single Hold-Out Designs. In each, we are evaluating on a single period and/or single set of assets. In later sections we'll explore Multiple Hold-Out Designs.

Note: The classic 12-2 momentum signal ala Jegadeesh and Titmam (1993) goes a step further and defines Dec2025 (t) momentum as returns Dec24 (t-12) through Oct25 (t-2), then tests on Jan26 and beyond

Outline

Evaluating the Signal with Future Returns

Toy Momentum Example

- Define asset-specific momentum during the training window as

$$\text{mom}_i^{\text{Train}}$$

- Construct the signal

$$s_i^{\text{Train}} = \begin{cases} +1 & \text{if } \text{mom}_i^{\text{Train}} > 0, \\ -1 & \text{if } \text{mom}_i^{\text{Train}} < 0, \\ 0 & \text{if } \text{mom}_i^{\text{Train}} = 0 \end{cases}$$

- Evaluate predictive ability: compare test-period returns

$$r_i^{\text{Test}} \mid s_i^{\text{Train}} = +1 \quad > \quad r_i^{\text{Test}} \mid s_i^{\text{Train}} = -1$$

Difference in Means

$$H_0: \ \mathbb{E}[r_i^{\text{Test}} \mid s_i^{\text{Train}} = +1] \leq \mathbb{E}[r_i^{\text{Test}} \mid s_i^{\text{Train}} = -1]$$

$$H_1: \ \mathbb{E}[r_i^{\text{Test}} \mid s_i^{\text{Train}} = +1] > \mathbb{E}[r_i^{\text{Test}} \mid s_i^{\text{Train}} = -1]$$

$$T = \frac{\frac{1}{N_{\text{Long}}} \sum_{i:s_i^{\text{Train}}=+1} r_i^{\text{Test}} - \frac{1}{N_{\text{Short}}} \sum_{i:s_i^{\text{Train}}=-1} r_i^{\text{Test}}}{\sqrt{\frac{\hat{\sigma}^2_{\text{Long}}}{N_{\text{Long}}} + \frac{\hat{\sigma}^2_{\text{Short}}}{N_{\text{Short}}}}}$$

$$\hat{\sigma}^2_{\text{Long}} = \frac{1}{N_{\text{Long}} - 1} \sum_{i:s_i^{\text{Train}}=+1} \left(r_i^{\text{Test}} - \frac{1}{N_{\text{Long}}} \sum_{j:s_j^{\text{Train}}=+1} r_j^{\text{Test}} \right)^2$$

- Define σ^2_{Short} analogously
- If reject the Null, then this is evidence in favor of the signal's power

Distributional Comparisons

Compare Test-period return distributions conditional on the Train-period signal

- **Signal frequencies and coverage:**

$$N_+, \; N_0, \; N_-, \qquad \Pr(s^{Train} = +1), \; \Pr(s^{Train} = 0), \; \Pr(s^{Train} = -1)$$

$$\text{Coverage} = \Pr(s^{Train} \neq 0) = \frac{N_+ + N_-}{N}$$

- **Conditional return summaries:** compare empirical moments and quantiles across

$$r^{Test} \mid s^{Train} = +1, \qquad r^{Test} \mid s^{Train} = 0, \qquad r^{Test} \mid s^{Train} = -1$$

  for each signal category with sufficient observations.
- **Visual diagnostics:**
  - Compare boxplots or histograms across signal groups.
  - Overlay kernel densities to examine differences in location, dispersion, skewness, and tails.
- **Nonparametric distributional test:**
  - Kolmogorov–Smirnov: test whether two conditional return distributions differ anywhere in their empirical CDFs.

Information Coefficient

Cross-sectional linear association between signal and forward returns

$$\boxed{\mathrm{IC} = \mathrm{Corr}_i(s_i^{Train}, r_i^{Test}) = \frac{\widehat{\mathrm{Cov}}(s^{Train}, r^{Test})}{\hat{\sigma}_s^{Train}\hat{\sigma}_r^{Test}}}$$

- s_i^{Train} : signal for asset $i \in \{-1, 0, +1\}$
- r_i^{Test} : forward return in the testing period
- $\widehat{\mathrm{Cov}}(s^{Train}, r^{Test}) = \frac{1}{N}\sum_i (s_i^{Train} - \bar{s}^{Train})(r_i^{Test} - \bar{r}^{Test})$

Information Coefficient (Acted–Only)

Correlation restricted to assets with active signals ($s \in \{-1, +1\}$)

$$\boxed{IC^{\text{acted}} = \frac{(\bar{r}^{Test} \mid s^{Train} = +1) - (\bar{r}^{Test} \mid s^{Train} = -1)}{\hat{\sigma}_{r^{Test}}} \sqrt{p(1-p)}}$$

- $\bar{r}^{Test} \mid s^{Train} = +1,\ \bar{r}^{Test} \mid s^{Train} = -1$: mean test-period returns for long and short groups
- $\hat{\sigma}_r^{Test}$: standard deviation of testing period returns on the acted subset
- $p = \dfrac{N_+}{N_+ + N_-}$: proportion of longs in the acted subset
- Equivalent to the Pearson correlation between $\{s_i^{Train}, r_i^{Test}\}$ for $s_i^{Train} \in \{+1, -1\}$

Parametric correlation test

$$H_0 : IC = 0 \qquad H_A : IC \neq 0$$

$$t = \frac{\text{IC}\sqrt{N-2}}{\sqrt{1-\text{IC}^2}}, \qquad t \sim t_{N-2}.$$

- N: number of assets (use N_{act} for acted-only case)
- Assumes $r^{Test}|s^{Train}$ approximately Normal
- Equivalent to a difference-in-means test for binary signals

Resampling alternatives (more robust to non-normality and small N)

- **Bootstrap:** resample asset pairs $(s_i^{Train}, r_i^{Test})$, recompute IC, build Confidence Interval.
- **Permutation:** shuffle signal labels across assets to form an empirical null:
    - Recompute IC for each shuffle to build the null distribution.
    - Compare observed IC to this distribution $\Rightarrow$ empirical p-value.

Confusion Matrix

Long vs Not Long (i.e. Short or Neutral)

	$s^{Train} = +1$	$s^{Train} = \{-1, 0\}$
$r^{Test} > 0$	True Positive (TP)	False Negative (FN)
$r^{Test} < 0$	False Positive (FP)	True Negative (TN)

Each cell of the table is frequency of observations meeting that criterion.

- $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$
- $Precision = \frac{TP}{TP+FP}$
- $Recall = \frac{TP}{TP+FN}$
- $Specificity = \frac{TN}{TN+FP}$
- Type 1 Error $= \frac{FP}{FP+TN}$
- Type 2 Error $= \frac{FN}{TP+FN}$

Confusion Matrix

Q: Can you interpret each of the metrics?

	$s^{Train} = +1$	$s^{Train} = \{-1, 0\}$
$r^{Test} > 0$	True Positive (TP)	False Negative (FN)
$r^{Test} < 0$	False Positive (FP)	True Negative (TN)

Each cell of the table is frequency of observations meeting that criterion.

- $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$ (Fraction of all trades where signal correctly anticipated sign of test period return; how often I got the sign right)
- $Precision = \frac{TP}{TP+FP}$ (Among assets you went long, the fraction that had positive test period returns; when I go long, how often I'm right)
- $Recall = \frac{TP}{TP+FN}$ (Among all assets that had positive test period returns, the fraction signal labeled long; winner capture rate)
- $Specificity = \frac{TN}{TN+FP}$ (Among all assets that had negative test period returns, the fraction signal labeled short; avoid going long losers)
- Type 1 Error $= \frac{FP}{FP+TN}$ (Among all assets with negative test period returns, the fraction signal incorrectly labeled long; mistakenly go long)
- Type 2 Error $= \frac{FN}{TP+FN}$ (Among all assets with positive test period returns, the fraction signal incorrectly labeled as short; missed winners)

Confusion Matrix

Q: Can you create and interpret a Confusion Matrix for Shorts?

	$s^{Train} = -1$	$s^{Train} = \{+1, 0\}$
$r^{Test} < 0$	True Positive (TP)	False Negative (FN)
$r^{Test} > 0$	False Positive (FP)	True Negative (TN)

Each cell of the table is frequency of observations meeting that criterion.

- $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$ (Fraction of all trades where signal correctly anticipated sign of test period return; how often I got the sign right)
- $Precision = \frac{TP}{TP+FP}$ (Among assets you went short, the fraction that had negative test period returns; when I go short, how often I'm right)
- $Recall = \frac{TP}{TP+FN}$ (Among all assets that had negative test period returns, the fraction signal labeled short; winner capture rate)
- $Specificity = \frac{TN}{TN+FP}$ (Among all assets that had positive test period returns, the fraction signal labeled long; avoid going short losing trades)
- Type 1 Error $= \frac{FP}{FP+TN}$ (Among all assets with positive test period returns, the fraction signal incorrectly labeled short; mistakenly go short)
- Type 2 Error $= \frac{FN}{TP+FN}$ (Among all assets with negative test period returns, the fraction signal incorrectly labeled as long; missed winning trades)

Defining the Hit Rate

Q: What is the meaning of "Hit" Rate?

A: A measure of the directional accuracy between signal and future returns

$$\text{HitRate} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[s_i^{Train} r_i^{Test} > 0]$$

$$\text{HitRate}^{\text{acted}} = \frac{\sum_{i=1}^{N} \mathbf{1}[s_i^{Train} \neq 0]\mathbf{1}[s_i^{Train} r_i^{Test} > 0]}{\sum_{i=1}^{N} \mathbf{1}[s_i^{Train} \neq 0]}$$

Testing the Hit Rate

Does directional accuracy different than random guessing?

Large-sample z-test (Binomial Normal Approximation)

$$H_0 : p = 0.5 \qquad H_A : p \neq 0.5$$

$$z = \frac{\hat{p} - 0.5}{\sqrt{p_0(1-p_0)/N}}, \qquad p_0 = 0.5, \qquad z \sim N(0,1) \text{ for large } N$$

- $\hat{p}$ = observed hit rate (use $\text{HitRate}_{\text{acted}}$ because the $p_0 = 0.5$ benchmark applies naturally to binary long/short directional predictions).
- Under H_0, number of hits $K \sim \text{Binomial}(N, 0.5)$.
- Exact binomial test preferred for small N or low coverage.
- In time-series or panel settings, dependence may inflate significance; use HAC or block bootstrap as robustness checks.

Reject H_0 at 5% if $|z| > 1.96$. Always report coverage and effective sample size.

Outline

Cross Validation (CV)

Robustness comes from many hold-outs, not just one

- **Time Series CV (Walk-Forward)**
  - Train: Dec'24 – Nov'25 Test: Dec'25
  - Train: Jan'25 – Dec'25 Test: Jan'26
  - ...
- **Cross-Sectional k-Fold CV (Assets)**
  - Example (*illustrative*, $k = 3$):
    - Train: A,B,C,D,E,F Test: G,H
    - Train: A,B,C,D,G,H Test: E,F
    - Train: E,F,G,H Test: A,B,C,D
  - Test groups are disjoint; each asset appears once as test data
  - In practice, choose k so that N/k is large enough to evaluate IC, Hit Rate, and AUC
- Note: We can combine Time Series and Cross Section CV

Creating a Long–Short Signal Portfolio

Q: How form a long short portfolio to illustrate the efficacy of the signal?

Define the long–short portfolio return for test period t as

$$r_{LS,t}^{Test} = \frac{1}{N_t^{Long}} \sum_{i:s_{i,t}^{\text{Train}}=+1} r_{i,t}^{\text{Test}} - \frac{1}{N_t^{Short}} \sum_{i:s_{i,t}^{\text{Train}}=-1} r_{i,t}^{\text{Test}},$$

where N_t^{Long} and N_t^{Short} denote the number of long and short positions, and $s_{i,t}^{\text{Train}} \in \{-1, 0, +1\}$ is the signal generated from the training window preceding the test period t.

Assessing the Long–Short Signal Portfolio

- **Seasonality:** table (rows = months, columns = years, entries = mean $r_{LS,t}^{Test}$)
- **Empirical density:** compare distribution of $r_{LS,t}^{Test}$ to benchmark returns
- **Downside:** underwater plot and summary of drawdowns

Rolling Information Coefficient

Summarize cross-sectional IC per test window (both full and acted-only)

$$\boxed{\mathsf{IC}_t \;=\; \mathrm{Corr}_i\big(s_{i,t}^{Train},\; r_{i,t+h}^{Test}\big) \text{ and } \mathsf{IC}_t^{\text{acted}} \;=\; \mathrm{Corr}_{i:\, s_{i,t}^{Train}\neq 0}\big(s_{i,t}^{Train},\; r_{i,t+h}^{Test}\big)}$$

$$\overline{\mathsf{IC}} \;=\; \frac{1}{T}\sum_{t=1}^{T}\mathsf{IC}_t \qquad\qquad \overline{\mathsf{IC}}^{\text{acted}} \;=\; \frac{1}{T}\sum_{t=1}^{T}\mathsf{IC}_t^{\text{acted}}$$

- Newey–West t-stat for mean IC:

$$t_{\mathsf{NW}} \;=\; \frac{\overline{\mathsf{IC}}}{\sqrt{\widehat{Var}_{NW}(\overline{\mathsf{IC}})}} \quad \text{(use analogous formula for } \overline{\mathsf{IC}}^{\text{acted}}\text{).}$$

- Autocorrelation and half-life (use series $\{\mathsf{IC}_t\}$ or $\{\mathsf{IC}_t^{\text{acted}}\}$):

$$\rho_1 = \mathrm{corr}(\mathsf{IC}_t, \mathsf{IC}_{t+1}), \qquad \text{Half-life} = -\frac{\ln 2}{\ln|\rho_1|}.$$

Summarizing Rolling IC

Statistic	IC (full)	IC (acted-only)
Mean	$\overline{\mathrm{IC}}$	$\overline{\mathrm{IC}}^{\mathrm{acted}}$
Std. dev.	$\hat{\sigma}_{\mathrm{IC}}$	$\hat{\sigma}_{\mathrm{IC}}^{\mathrm{acted}}$
Newey–West t	t_{NW}	$t_{\mathrm{NW}}^{\mathrm{acted}}$
Fraction positive	$\Pr(\mathrm{IC}_t > 0)$	$\Pr(\mathrm{IC}_t^{\mathrm{acted}} > 0)$
Mean coverage	—	$\overline{\mathrm{Coverage}} = \frac{1}{T}\sum_t \frac{N_{\mathrm{act},t}}{N_t}$
Lag-1 autocorr	ρ_1	ρ_1^{acted}
Half-life (periods)	HL	$\mathrm{HL}^{\mathrm{acted}}$

Recommended visuals / diagnostics:

- Time series plot of IC_t and $\mathrm{IC}_t^{\mathrm{acted}}$ (overlay) with shaded NW CI.
- Histogram / density of $\{\mathrm{IC}_t\}$ and $\{\mathrm{IC}_t^{\mathrm{acted}}\}$; table of summary moments.
- Coverage time series and counts $N^{Long}, N^{Short}, N^{Neutral}$ per period.
- Bootstrap / permutation p-values for $\overline{\mathrm{IC}}$ when N_t^{acted} small or imbalanced.

Rolling Hit Rate

Time-series of directional accuracy (explicit handling of neutrals)

$$\mathrm{HitRate}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \mathbf{1}[s_{i,t}^{Train} r_{i,t}^{Test} > 0]$$

$$\mathrm{HitRate}_t^{\mathrm{acted}} = \frac{\sum_{i=1}^{N_t} \mathbf{1}[s_{i,t}^{Train} \neq 0] \mathbf{1}[s_{i,t}^{Train} r_{i,t}^{Test} > 0]}{\sum_{i=1}^{N_t} \mathbf{1}[s_{i,t}^{Train} \neq 0]}$$

Rolling Hit Rate - Aggregation and Inference

- **Time-series aggregation:**

$$\overline{\text{HitRate}} = \frac{1}{T}\sum_{t=1}^{T} \text{HitRate}_t, \qquad \overline{\text{HitRate}}^{\text{acted}} = \frac{1}{T}\sum_{t=1}^{T} \text{HitRate}_t^{\text{acted}}$$

- **Preferred null for directional accuracy:** conduct inference on $\overline{\text{HitRate}}^{\text{acted}}$ and test

$$H_0 : \mathbb{E}\left[\text{HitRate}_t^{\text{acted}}\right] = 0.5,$$

  where 0.5 represents random directional accuracy.
- Neutral positions are excluded because counting them as correct can mechanically inflate the measured hit rate without reflecting successful directional predictions.

Rolling Hit Rate - Aggregation and Inference cont'd

- The sequence $\{\text{HitRate}_t^{\text{acted}}\}_{t=1}^{T}$ may be **serially correlated** and/or **heteroscedastic**. Newey-West adjusts the estimator of the sample mean for both.
- **Newey–West variance:** let $x_t = \text{HitRate}_t^{\text{acted}}$, $\bar{x} = \overline{\text{HitRate}}^{\text{acted}}$
  The Newey–West estimate of the variance of the sample mean is

$$\widehat{\mathrm{Var}}_{\mathrm{NW}}(\bar{x}) = \frac{1}{T}\left[\hat{\gamma}_0 + 2\sum_{\ell=1}^{L}\left(1 - \frac{\ell}{L+1}\right)\hat{\gamma}_\ell\right],$$

  where $\hat{\gamma}_\ell = \frac{1}{T}\sum_{t=\ell+1}^{T}(x_t - \bar{x})(x_{t-\ell} - \bar{x})$ is the estimated autocovariance at lag ℓ.
- **Newey–West test for mean hit rate:**

$$t_{\mathrm{NW}} = \frac{\overline{\text{HitRate}}^{\text{acted}} - 0.5}{\sqrt{\widehat{\mathrm{Var}}_{\mathrm{NW}}\left(\overline{\text{HitRate}}^{\text{acted}}\right)}}.$$

- The truncation lag L determines how many lagged autocovariances are incorporated; select via rule-of-thumb/data-driven.

Rolling Hit Rate - Aggregation and Inference cont'd

- **Per-period exact test:** for each period t, let $N_t^{\text{acted}} = \sum_{i=1}^{N_t} \mathbf{1}[s_{i,t}^{\text{Train}} \neq 0]$, and compute a Binomial$(N_t^{\text{acted}}, 0.5)$ p-value to flag unusually strong or weak months.
- **Resampling robustness:** block bootstrap over t (to preserve serial dependence), or permutation tests that shuffle realized returns across assets within each t, provide empirical p-values when serial correlation or small counts are concerns.

Rolling Hit Rate — Persistence Diagnostics

Is directional accuracy persistent or transitory?

Persistence diagnostic	Definition / Interpretation
Fraction of positive periods	$\Pr\left(\mathsf{HitRate}_t^{\mathsf{acted}} > 0.5\right)$ Frequency with which directional accuracy exceeds random guessing
Lag-1 autocorrelation	$\rho_1^{\mathsf{HitRate}} = \operatorname{corr}\left(\mathsf{HitRate}_t^{\mathsf{acted}}, \mathsf{HitRate}_{t+1}^{\mathsf{acted}}\right)$ Short-run persistence of directional accuracy
Half-life (periods)	$\mathsf{HalfLife}_{\mathsf{HitRate}} = -\dfrac{\ln 2}{\ln\left\lvert\rho_1^{\mathsf{HitRate}}\right\rvert}$ Time required for abnormal accuracy to decay by 50%
Mean coverage	$\overline{\mathsf{Coverage}} = \dfrac{1}{T}\sum_{t=1}^{T} \dfrac{N_t^{\mathsf{acted}}}{N_t}$ Fraction of assets with non-zero positions

Hit Rate — Diagnostics & Visuals

Focus on acted signal; show stability, significance, and participation

Recommended plots / diagnostics

- Time series: overlay $\text{HitRate}_t^{\text{acted}}$ and Coverage_t (dual axis); mark months with Binomial p-value < 0.05.
- Rolling estimate: 6–12 month rolling mean of $\text{HitRate}_t^{\text{acted}}$ with Newey–West CI shading.
- Heatmap of per-period Binomial p-values (months $\times$ years) to flag exceptional windows.
- Distribution: histogram / density of $\{\text{HitRate}_t^{\text{acted}}\}$ with block-bootstrap CI for $\overline{\text{HitRate}}^{\text{acted}}$.
- Scatter: $\text{HitRate}_t^{\text{acted}}$ vs $\text{IC}_t^{\text{acted}}$; annotate correlation and points with low coverage.
- Stability: plot lag-1 autocorr and implied half-life (annotate numeric value).

Outline

Anatomy of a Strategy

Strategy converts signal into investments

1. Timing: when do we enter and exit?
2. Position Sizing: how much do we invest in each asset?

Outline

Position Timing

Converting raw signal s into a traded position s^{pos}

Raw ML signal:

$$s_{i,t} \in \{-1, 0, +1\} \qquad \text{(what the model suggests at time } t\text{)}$$

Example:

$$s_{i,\text{Jan}} = +1, \qquad s_{i,\text{Feb}} = 0$$

- Model says: *Long in January, neutral in February*
- Naively: enter in Jan, exit immediately in Feb
- In practice: the trading strategy may prefer to *hold longer*

Solution: introduce a *traded signal*

$$s^{pos}_{i,t} \in \{-1, 0, +1\}$$

- $s^{pos}_{i,t}$ converts raw signal $s_{i,t}$ into an actual position

Interpreting $s_{i,t}^{pos}$

The trading layer on top of the signal

- The signal $s_{i,t}$ is evaluated at each time t
- Once a position is opened, it is typically *held*
- The position is closed only when an exit rule is triggered

Common exit trigger types:

- **Time-based:** exit after a fixed holding period
- **Asset-based:** exit due to asset-specific performance or risk
- **State-based:** exit due to changes in market or portfolio conditions

Appendix: Formal View of Position Timing

A state-based representation of how signals become positions

$$\boxed{(s_t^{\text{pos}},\, h_t,\, \tau_t,\, e_t) \;=\; \mathcal{G}\big(s_t,\, h_{t-1},\, \tau_{t-1},\, X_t,\, Z_t;\, \Theta\big)}$$

- $s_t = [s_{1,t}, \dots, s_{N,t}]^\top$: raw signals produced by the model at time t
- $s_t^{\text{pos}} \in \{-1, 0, +1\}^N$: positions held over $[t, t+1)$
- $h_{t-1} \in \{0, 1\}^N$: indicators for whether a position is already open
- $\tau_{t-1} \in \mathbb{N}^N$: holding durations entering time t
- e_t: exit indicator / reason (only nonzero when an exit occurs)
- Θ: timing and exit parameters (holding limits, stops, filters)

Outline

Raw Weights: general mapping

Map timing-adjusted signals into signed pre-normalized weights

Let $s_t^{pos} = [s_{1,t}^{pos}, \dots, s_{N,t}^{pos}]^\top$ denote the timing-adjusted signal. We define the *raw* weight vector at time t as a (possibly elementwise) mapping:

$$\boxed{w_t^{\text{raw}} \; = \; \Phi\big(s_t^{pos},\, X_t,\, Z_t;\, \Theta_\Phi\big)}$$

where

- $X_t = \{X_{i,t}\}_{i=1}^N$ are asset-level features (e.g. returns, vol, forecasts),
- Z_t are contextual / cross-sectional features (market vol, cross-section medians, universe set),
- Θ_Φ: parameters of the sizing map.

Properties / conventions:

- w_t^{raw} is a signed, pre-normalized weight (an intermediate exposure); it does not by itself satisfy portfolio constraints and must be transformed by scaling/optimization to produce tradable weights.
- Typically $w_{i,t}^{\text{raw}} = 0$ when $s_{i,t}^{pos} = 0$ (no action).
- The mapping Φ should be deterministic for reproducibility in backtests.

Raw Example 1: Equal-weight per acted asset

Simple, non-parametric sizing: equal exposure to every active name

Let the acted set be $\mathcal{H}_t = \{i : s_{i,t}^{pos} \neq 0\}$ and $N_t^{\text{acted}} = |\mathcal{H}_t|$. A minimal raw weighting is

$$w_{i,t}^{\text{raw}} = s_{i,t}^{pos}.$$

This expresses direction only. A simple cross-sectionally normalized (still pre-portfolio) version is

$$w_{i,t}^{\text{raw}} = \frac{s_{i,t}^{pos}}{N_t^{\text{acted}}}.$$

Notes:

- Very low model risk and easy interpretation.
- Appropriate when the strategy emphasizes selection over sizing.
- Downstream scaling converts these into investable portfolio weights.

Raw Example 2: Volatility-scaled (risk-aware) raw weights

Scale directional intent by inverse realized volatility

Let $\sigma_{i,t}$ be an estimate of asset i's volatility (e.g. 20-day realized). A common raw mapping is

$$w_{i,t}^{\text{raw}} = s_{i,t}^{pos} \cdot \frac{1}{\sigma_{i,t}}.$$

Practical adjustment:

$$w_{i,t}^{\text{raw}} \leftarrow \text{clip}\left(w_{i,t}^{\text{raw}}, \; -w_{\text{max}}^{\text{raw}}, \; w_{\text{max}}^{\text{raw}}\right)$$

to prevent extreme pre-weights from very low $\sigma_{i,t}$.
Notes:

- Equalizes per-name risk contribution *before* portfolio normalization.
- Improves stability of downstream scaling in heterogeneous universes.
- Conceptually distinct from full portfolio-level risk parity.

Raw Example 3: Forecast-magnitude raw weights

Size positions by expected return magnitude

Suppose the model produces a one-step-ahead return forecast $\hat{r}_{i,t+1}$. Define the raw weight

$$w_{i,t}^{\text{raw}} \;=\; s_{i,t}^{pos} \cdot g\big(\hat{r}_{i,t+1}\big)\,, \qquad g(\hat{r}) \geq 0.$$

Common choices for $g(\cdot)$:

$$g_1(\hat{r}) = |\hat{r}|, \qquad g_2(\hat{r}) = \frac{|\hat{r}|}{\sigma_{i,t}}, \qquad g_3(\hat{r}) = \min(|\hat{r}|,\, c)\,.$$

Notes:

- Direction is determined by $s_{i,t}^{pos}$.
- Magnitude reflects the model's expected return signal.
- Most natural when forecasts are well-calibrated and comparable cross-sectionally.

Raw Example 4: Confidence-based raw weights

Size positions by model confidence rather than return magnitude

Suppose the model produces a nonnegative confidence score $\pi_{i,t} \geq 0$ (e.g. class probability, margin, or posterior state probability). Define

$$w_{i,t}^{\text{raw}} \;=\; s_{i,t}^{pos} \cdot \pi_{i,t}^{\alpha}, \qquad \alpha \in (0,1].$$

Notes:

- Direction is determined by $s_{i,t}^{pos}$.
- $\pi_{i,t}$ controls position magnitude via confidence in the directional call.
- More robust than return-based sizing when magnitude forecasts are noisy.

Scaled Weights (w^{scaled})

Make raw weights investable via normalization and mechanical constraints

$$\boxed{w_t^{\text{scaled}} = \mathcal{N}\big(w_t^{\mathsf{raw}};\ \mathcal{C}_t\big) = \Pi_{\mathcal{C}_t}\big(\mathcal{S}(w_t^{\mathsf{raw}})\big)}$$

where

- $\mathcal{S}(\cdot)$ is a normalization operator (e.g. gross or net scaling),
- $\Pi_{\mathcal{C}_t}(\cdot)$ enforces feasibility with respect to a set of mechanical constraints $\mathcal{C}_t$,
- $\mathcal{C}_t$ may include: budget constraints, leverage limits, single-name or sector caps, long/short splits, liquidity rules.

Common normalization choices:

$$\mathcal{S}_{L1}(w) = \frac{w}{\sum_i |w_i|}$$

Long–short (gross exposure)

$$\mathcal{S}_{\text{net}}(w) = \frac{w}{\sum_i w_i}$$

Long-only (net exposure)

Normalization and constraint rules should be consistent across strategies so the portfolio layer can combine them meaningfully. Scaling of w^{raw} may set *relative* exposure across assets; w^{scaled} enforces *absolute* budget and feasibility constraints.

Example: L1 normalization with long/short budgets

Define total raw long and short mass:

$$G_t^+ = \sum_{i:w_{i,t}^{\text{raw}}>0} |w_{i,t}^{\text{raw}}|, \qquad G_t^- = \sum_{i:w_{i,t}^{\text{raw}}<0} |w_{i,t}^{\text{raw}}|$$

Choose long and short budgets $B_L \geq 0$ and $B_S \geq 0$. Define scaled weights:

$$w_{i,t}^{\text{scaled}} = \begin{cases} B_L \dfrac{|w_{i,t}^{\text{raw}}|}{G_t^+}, & w_{i,t}^{\text{raw}} > 0, \\ -B_S \dfrac{|w_{i,t}^{\text{raw}}|}{G_t^-}, & w_{i,t}^{\text{raw}} < 0, \\ 0, & w_{i,t}^{\text{raw}} = 0 \end{cases}$$

Properties:

- Gross exposure $= B_L + B_S$; net exposure $= B_L - B_S$
- Preserves relative raw exposures within each side.
- Single-name or sector caps can be enforced by clipping and re-normalizing per side.

Example: Volatility-aware scaling

Normalize raw weights to control ex-ante volatility under diagonal risk

Start from raw weights $w_{i,t}^{\text{raw}}$ and per-asset volatility estimates $\sigma_{i,t}$. Define volatility-adjusted pre-weights:

$$\tilde{w}_{i,t} = \frac{w_{i,t}^{\text{raw}}}{\sigma_{i,t}}.$$

Choose a gross exposure target $G > 0$ and scale via L_1 normalization:

$$w_{i,t}^{\text{scaled}} = \frac{\tilde{w}_{i,t}}{\sum_j |\tilde{w}_{j,t}|} \cdot G.$$

Interpretation:

- Preserves directional intent and relative raw strength.
- Equalizes exposure to per-asset volatility (ignoring correlations).
- Controls portfolio gross exposure mechanically.

G sets the target gross exposure: $\sum_i |w_{i,t}^{\text{scaled}}| = G$.
a portfolio-level optimization step and belongs in the w^p layer.

Outline

Strategy Evaluation

Standalone performance of a fully specified trading rule

- Input: *fixed* scaled weights $w_{i,t}$ (direction + exposure / scaling already applied).
- Output: single-strategy return series $\{r_t^{\text{strat}}\}$ to be evaluated before any portfolio-level allocation.
- Scope: performance of one strategy at a time; no cross-strategy optimization or global constraints here.

From Scaled Weights to Strategy Returns

$$r_{t+1}^{\text{strat}} = \sum_{i=1}^{N} w_{i,t}^{scaled}\, r_{i,t+1}$$

where

- $w_{i,t}^{scaled}$: scaled weights or exposures at time
- $r_{i,t+1}$: realized simple asset returns over $[t, t+1)$

Include transaction costs / slippage (example):

$$r_{t+1}^{\text{net}} = \sum_{i} w_{i,t}^{scaled} r_{i,t+1} - c \sum_{i} \left| w_{i,t} - w_{i,t-1} \right|$$

where c is per-unit round-trip cost (or a function of turnover).

Notes

- Weights are *ex-ante*; returns are *realized*; align timestamps carefully.
- If your scaling uses gross exposure G or volatility target, it should already be embedded in $w_{i,t}$.

Performance Metrics: Four Categories

The following are a small sample of performance metrics available

1. **Absolute Return Measures — "Did it make money?"**
   - Total return (cumulative)
   - CAGR: $\mathrm{CAGR} = \left(\prod_t (1 + r_t^{\mathrm{strat}})\right)^{\frac{T_{\mathrm{yr}}}{T}} - 1$
   - Return moments: mean, median, skewness, kurtosis
   - Drawdown series: $\mathrm{DD}_t = 1 - \dfrac{(1 + R_t)}{\max_{s \le t}(1 + R_s)}$
2. **Risk Measures — "How volatile / tail-risky?"**
   - Annualized volatility: $\sigma_{\mathrm{ann}} = \sqrt{f}\,\mathrm{std}(r_t)$ where f = periods/yr
   - Downside volatility; semi-deviation
   - VaR / CVaR (e.g., 95%): empirical or parametric
   - Maximum drawdown (depth and length)
   - Turnover: $\sum_t \sum_i |w_{i,t} - w_{i,t-1}|$

Performance Metrics (cont.)

Relative, statistical, and practical metrics

3. Risk-Adjusted Measures — "Was return worth the risk?"

- Sharpe ratio: $\mathrm{SR} = \dfrac{\bar{r}_{\mathrm{ann}} - r_f}{\sigma_{\mathrm{ann}}}$
- Sortino ratio (use downside deviation)
- Calmar ratio: $\dfrac{\text{CAGR}}{\text{Max Drawdown}}$
- Information Ratio (when benchmarked): $\dfrac{\overline{r - \text{bench}}}{\sigma(r - \text{bench})}$

4. Statistical Reliability Measures — "Skill or noise?"

- t-stat of mean return (adjust for autocorrelation / heteroskedasticity)
- Newey–West adjusted Sharpe / standard errors
- Bootstrap confidence intervals for CAGR, Sharpe, MaxDrawdown
- Deflated Sharpe (account for multiple trials / backtest overfitting)
- Subsample / rolling stability (e.g., rolling Sharpe, rolling IC)

Outline

Portfolios

A portfolio is a combination of strategies subject to global constraints

- Each *strategy* $k = 1, \ldots, K$ produces a return stream $r_t^{\text{strat},k}$.
- A *portfolio* combines these strategies using weights $u_t^{\text{strat},k}$
- Portfolio return r_t^p is a weighted combination of $\{r_t^{\text{strat},k}\}$
- Strategy weights u_t^{strat} are chosen to optimize an objective (e.g. risk, return, or risk-adjusted performance) subject to *strategy-level* and *asset-level* constraints
- Although optimization occurs at the strategy level, the portfolio ultimately holds *assets*. Asset weights w_t^p are implied by:
  - each strategy's internal asset weights, and
  - the portfolio's allocation across strategies

Portfolio Components

Strategy construction (asset level):

$$r_t^{\text{strat},k} = \sum_{i=1}^{N} w_{i,t}^{\text{scaled},k} \, r_{i,t}, \qquad k = 1, \ldots, K$$

Portfolio construction (strategy level):

$$r_t^{p} = \sum_{k=1}^{K} u_t^{\text{strat},k} \, r_t^{\text{strat},k}$$

Implied portfolio asset weights:

$$w_{i,t}^{p} = \sum_{k=1}^{K} u_t^{\text{strat},k} \, w_{i,t}^{\text{scaled},k}$$

Equivalently, in matrix form:

$$w_t^{p} = W_t^{\text{scaled}} \, u_t^{\text{strat}}$$

Portfolio Formation — Optimization

$$u_t^* \;=\; \arg\max_u \mathbb{E}_t\left[U\left(\sum_{k=1}^{K} u_t^k \, r_t^{\text{strat},k} \right) \right] \;-\; \mathbb{E}_t\big[\ell\big(\Delta w_t^p(u)\big)\big]$$

$$\begin{aligned} \text{s.t.}\quad & A\, w_t^p \le b, && A_{\text{eq}}\, w_t^p = b_{\text{eq}} \\ & C\, u_t \le d, && C_{\text{eq}}\, u_t = d_{\text{eq}} \\ & w_t^p = W_t^{\text{scaled}}\, u_t \end{aligned}$$

- $U(\cdot)$: portfolio objective (e.g. variance, mean–variance, expected utility).
- $\ell(\cdot)$: transaction cost function; $\Delta w_t^p(u) = w_t^p - w_{t-1}^p$.
- A, b: asset-level constraint matrices and bounds ($A \in \mathbb{R}^{R_1 \times N}$).
- C, d: strategy-level constraint matrices and bounds ($C \in \mathbb{R}^{R_2 \times K}$).
- W_t^{scaled}: matrix of strategy asset weights ($N \times K$), mapping strategy allocations to asset weights.

Strategy-level Constraints

$K = 2$ strategies; $R_2 = 2$ inequality constraints and 1 equality constraint

- *No shorting of strategies:*

$$u_t^{\text{strat},k} \geq 0, \qquad k = 1, 2$$

$$\underbrace{\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}}_{C \in \mathbb{R}^{R_2 \times K}} \underbrace{\begin{bmatrix} u_t^{\text{strat},1} \\ u_t^{\text{strat},2} \end{bmatrix}}_{u_t} \leq \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{d}$$

- *Fully invested among strategies (budget constraint):*

$$\sum_{k=1}^{2} u_t^{\text{strat},k} = 1$$

$$\underbrace{\begin{bmatrix} 1 & 1 \end{bmatrix}}_{C_{\mathrm{eq}} \in \mathbb{R}^{1 \times K}} \; u_t \; = \; \underbrace{\begin{bmatrix} 1 \end{bmatrix}}_{d_{\mathrm{eq}}}$$

Asset-level Constraints

Individual position limits

- *Maximum individual asset weight:*

$$w_{i,t}^{p} \leq 5\%, \qquad i = 1, \ldots, N$$

$$\underbrace{I_N}_{A_1 \in \mathbb{R}^{R_{1,1} \times N}} \; \underbrace{w_t^p}_{\in \mathbb{R}^N} \leq \underbrace{0.05 \cdot \mathbf{1}_N}_{b_1 \in \mathbb{R}^{R_{1,1}}}$$

Asset-level Constraints (cont'd)

Sector exposure limits

- *Sector exposure limits:*
  Define a sector–asset mapping matrix

$$S \in \{0,1\}^{M\times N}, \qquad S_{m,i} = 1 \text{ if asset } i \text{ belongs to sector } m.$$

  Then sector-level constraints can be written as:

$$S\, w_t^p \leq 0.20 \cdot \mathbf{1}_M$$

$$\underbrace{S}_{A_2 \in \mathbb{R}^{R_{1,2}\times N}} \; \underbrace{w_t^p}_{\in \mathbb{R}^N} \leq \underbrace{0.20 \cdot \mathbf{1}_M}_{b_2 \in \mathbb{R}^{R_{1,2}}}$$

Summary:

$$A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \in \mathbb{R}^{R_1\times N}, A \in R^{M\times N} \qquad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}, \qquad Aw_t^p \leq b$$

- Classic mean–variance optimization (MVO), in the Markowitz tradition, produces portfolio weights that trade off expected return against risk using an expected return vector and a covariance matrix.
- In this course framework, MVO is a *building block*:
  - within a *strategy*, MVO may be used to allocate across assets;
  - at the *portfolio level*, MVO may be used to allocate across strategies.
- Importantly, MVO itself does not specify its inputs. Expected returns and covariances must be supplied by upstream signals, models, or estimators.

Mean–Variance Optimization (MVO) cont'd

Practical considerations and implementation details

- When the investable universe is large, MVO can be numerically unstable due to estimation error in large covariance matrices. A common practical response is to *pre–filter* the universe using a signal.
- For example, assets may be ranked by forecasted Sharpe ratio and a selection rule applied:

$$a_i = \begin{cases} 1, & \text{asset selected for optimization,} \\ 0, & \text{asset excluded.} \end{cases}$$

  Optimization is then performed only over assets with $a_i = 1$.
- In practice, strategy construction often proceeds in stages:
  1. Compute provisional weights (e.g. minimum variance or mean–variance) on the filtered universe.
  2. Rescale or re–optimize to enforce global constraints such as full investment.
  3. Impose additional asset–level constraints (e.g. position limits, sector limits) to obtain final portfolio weights w^p.